# Zilai Li

*Independent Researcher & Software Engineer*

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## SUMMARY

Independent researcher and software engineer focusing on efficient generative modeling, fast ODE/SDE solvers for diffusion models, training-free diffusion distillation enhancement, and lightweight semantic watermark extraction. Contributor to Hugging Face Diffusers with interactive model playgrounds hosted on Hugging Face Spaces (`coralLight`).

## EDUCATION

**University of Nottingham** — *M.Sc. in Computer Science (With Merit)* **2021 – 2022**
Jubilee Campus · Nottingham, England, GB

**China University of Geosciences, Beijing (CUGB)** — *B.Eng. in Computer Science and Technology* **2016 – 2020**
School of Information Engineering · Beijing, Beijing, CN

## RESEARCH & PUBLICATIONS

- **RAIN: Region-Aware Inversion Network for Semantic Watermark Extraction**
  **Zilai Li — BMVC 2026 Workshop on Trustworthy Visual Computing (TrustVis), Accepted for Oral Presentation** (arXiv:2609.14856 | OpenReview | GitHub)
  Proposed a prompt-free, U-Net-free, dual-branch NAFNet architecture for one-step Gaussian-Shading semantic watermark extraction. Decomposes high-SNR endpoint statistic recovery into concurrent anchor ($F_0$) and residual/timestep ($F_1, F_2$) branches across separate CUDA streams, achieving **47.4× fewer FLOPs** (`14.394 GFLOPs` vs. `~680 GFLOPs`), **56× fewer parameters** (`15.468M`), and **1.39 ms/image** latency (**22.5×–26.8× speedup** over OSI and FARI) while matching 50-step DDIM Inversion accuracy (`99.99%`) with strong robustness across 10 distortions.

- **F-scheduler: Illuminating the Free-Lunch Design Space for Fast Sampling of Diffusion Models**
  **Zilai Li**, Lujia Bai — **Under Review at Transactions on Machine Learning Research (TMLR)** (arXiv:2510.02390 | GitHub | HF Space Demo)
  Designed a training-free time and Free-U skip-connection schedule for few-step ($< 10$) diffusion ODE solvers (`UniPC`, `DPM++ 2M`) that preserves terminal KL-regularized latent variance for the $\beta$-VAE decoder. Samples `1024×1024` (SDXL) images in **6 steps** and `512×512` (SD 1.5) images in **5 steps**, outperforming SOTA diffusion distillation models (e.g., DMD2 LoRA) and 20-step DPM++ 2M in FID and visual fidelity.

- **EMPURPLE: A Free Lunch for Diffusion Distillation based on the Information Bottleneck**
  **Zilai Li**, Lujia Bai — **Working Paper / Preprint** (arXiv:2607.04276 | GitHub | HF Space Demo)
  Analyzed mode collapse in distilled diffusion models through a PAC-style generalization bound and the Information Bottleneck principle, showing that aggressive early-step velocity field redirection enlarges the train–test gap and shifts intermediate noisy latent distributions. Proposed a model-agnostic, training-free latent recycling method that improves FID by **7%–20%** across DMD2 (**+20% FID**), Hyper-SD, FlashSD, and SDXL-Lightning.

## EXPERIENCE & OPEN-SOURCE CONTRIBUTIONS

**Independent Researcher** — *Generative AI & Computer Vision* **2022 – Present**

- Authored independent research on diffusion ODE solvers (`F-scheduler`), distillation diversity (`EMPURPLE`), and semantic watermark extraction (`RAIN`, BMVC 2026 TrustVis Oral).
- **Hugging Face Diffusers Contributor:** Merged PR #14011 fixing custom timestep shifting in `FlowMatchEulerDiscreteScheduler.set_timesteps` when `shift != 1`.
- **Interactive Hugging Face Spaces (`coralLight`):** Built and deployed live Gradio demos including `DMD2vsF-Scheduler`, `EMPURPLE`, `F-scheduler-4k`, `F-scheduler-DreamshaperXL`, and `F-scheduler-Animation`.

**Software Engineer** — *Full-Time Industry Software Development* **2023 – Present**

- Designed and maintained production client architectures, MVVM reactive state pipelines, high-frame-rate UI/animation systems, and multi-GPU deep learning evaluation benchmarks.

## TECHNICAL SKILLS

- **Generative AI & Vision:** Diffusion Models (SD 1.5, SD v2.1, SDXL, DiT), Fast ODE/SDE Solvers (UniPC, DPM++ 2M, DDIM, Flow Matching), Diffusion Distillation (DMD2, LCM, Hyper-SD, SDXL-Lightning), Semantic Watermarking (Gaussian Shading)
- **Frameworks & Engineering:** PyTorch, Hugging Face Diffusers, xFormers, BasicSR, NAFNet, Multi-Stream CUDA Parallelism, Gradio, Python, Swift/iOS, C/C++, Git, Linux, Docker